# Gears, motor and first power

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Compare a gear-based prediction with a supervised first powered test.

**SUCCESS CRITERIA - I can...**

1. Check gear mesh and secure connections before power.
2. Calculate ideal wheel speed from measured or supplied rpm.
3. Document first-test evidence and one revision.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “circumference” and “gear ratio” to compare your model or result. Sentence starters are optional.

**circumference**: distance around the edge of a circle

**gear ratio**: a comparison of gear tooth counts or rotation rates

**assumption**: a condition accepted for using a model

## 01 Bring back an earlier idea

At unchanged wheel rpm, circumference becomes 3 times as large.
**Compare** ideal travel speed and state one real-world limitation.

**Word help:** ideal = using stated simplifying conditions

**Sentence starter:** Both ... . They differ in ... because ... .

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** rotation rate = turns/Δt
v = circumference × rotation rate.
Turns per second, no slip. Get first-power approval; secure car and keep loose items clear. Disconnect before adjusting gears.

**Worked reasoning:** Supplied wheel tests show 4 rotations in 2 s. With circumference 0.20 m and no slip, predicted speed is(4/2)×0.20=0.40 m/s. Ground speed may differ if wheels slip.

**Your turn: Draw** one wheel rotation and the predicted travel distance.
**Identify** the no-slip assumption.

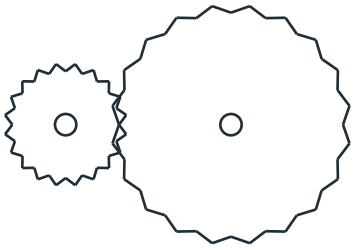
**Sentence starter:** I predict ... . My model supports this because ... .

**Marzano check - circle:** ① ② ③

## 03 Look and explain

**Sketch** your driver and driven gears.
**Label** tooth counts and identify the driven axle. Note any rubbing or loose mesh before power.

**Sentence starter:** The ... in my model represents ... . Its label shows ... .




**Marzano check - circle:** ① ② ③

## 04 Build the solution

A supplied motor turns 300 rpm through a 10-tooth driver and 30-tooth driven gear.
**Use** the supplied wheel rate 100 rpm and circumference .18 m.
**Find** ideal speed=.18×100÷60.

**Word help:** ideal = using stated simplifying conditions

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

**Marzano check - circle:** ① ② ③

# Gears, motor and first power

**antocciphysics**



Continue down each column.

Sentence starters are optional. You may plan in any language.

## 05 Use the idea

The actual car is slower than the prediction.
**Name** two assumptions to check and one measurement that could separate their effects.

**Sentence starter:** I predict … . My model supports this because … .

**Marzano check - circle:** 1 2 3

## 06 Choose a method

A consistently drawn magnetic-field map has 4 times as many field lines through the same small perpendicular area at A as at B. What does the denser spacing indicate?

**Sentence starter:** The … in my model represents … . Its label shows … .

**Marzano check - circle:** 1 2 3

## 07 Exit check

**Record** teacher approval, first-test observation and one next action.

**Sentence starter:** I notice … . This supports … because … .

**Marzano check - circle:** 1 2 3

## 08 STRETCH: Apply it another way

A car's wheels predict 0.40 m/s but measured ground speed is 0.30 m/s.
**Name** two hypotheses and a measurement to distinguish them.

**Sentence starter:** I predict … . My model supports this because … .

**Marzano check - circle:** 1 2 3